FINA 4103 midterm

1 · Papers and people

three reasons for a spread order processing (Demsetz), inventory risk (Smidt, Stoll), and information (Glosten–Milgrom); only the last one reverses in the price.
Garman 1976 JFE coined "microstructure"; orders as a Poisson process; two models, dealership and auction.
Demsetz 1968 QJE the spread is the price of immediacy: order-processing cost and time value.
Smidt 1971 FAJ the spread is inventory cost.
Stoll 1978 JF the spread pays for the price risk of the dealer's inventory.
Glosten & Milgrom 1985 JFE the spread is information friction: the maker loses to a trader who knows more.
Zhu 2014 RFS "Do Dark Pools Harm Price Discovery?" No: informed traders are rationed in the dark, leave for lit venues, and the lit price improves (a monopolist informed trader may differ).
Boulatov & George 2013 RFS hidden orders benefit the market overall.
Cao et al. 2022 JFE the share of immediate-or-cancel orders grew through the 2000s.
Lewis 2014 Flash Boys the direct data feed outruns the consolidated feed, so a quote vanishes as an order arrives.
Park, Suen & Wan 2022 JFM Hong Kong close sniping: sell orders at 4:09:57 drove the indicative close from 37 to 33 dollars.
Chen et al. 2024 JF "The Dark Side of Circuit Breakers": a halt makes the market more volatile and more fragile.
Budish, Cramton & Shim 2015 QJE the frequent batch auction, proposed against the high-frequency arms race.
Budish, Lee & Shim 2023 JPE exchange fees fall to negligible, because every venue sells the same execution (Bertrand).
Li, Ye & Zheng 2022 JFE "Refusing the best price?" New York Stock Exchange order-type data: 57% of orders refuse routing; races, price improvement.
Aquilina et al. 2020 the London Stock Exchange platform circuit; 6% of orders arrive at a gateway first but are sequenced behind a rival.
Basel CBS 2013 the post-crisis central-counterparty framework.
Stoll 2000 JF "Friction": tests Roll's estimator on 61 trading days; it recovers 0.0381 of the 0.079 quoted on the NYSE and 0.1115 of 0.129 on Nasdaq, so it understates the quote because processing and inventory cost never reverse.
Roll 1984 JF bid-ask bounce; the spread read from the negative autocovariance of price changes.
Kyle 1985 ECTA a strategic insider trades against noise; the size of the order he submits reveals part of what he knows and is the illiquidity the market charges.
named results asset pricing owns Black–Scholes–Merton, corporate finance owns Modigliani–Miller, and market microstructure owns this course.

2 · Statutes and rule numbers

Securities Exchange Act 1934 Rule 12(f) grants unlisted trading privileges: a stock listed on one exchange trades on all, so listing and trading are separate decisions. Apple lists only on Nasdaq and trades everywhere; BATS runs no listing service.
national market system the project from the mid-1970s to 2007 that made the 16 United States exchanges behave as one.
Regulation NMS Securities and Exchange Commission (SEC) rule, June 2005; four numbered rules.
Rule 611 order protection: no trade-through; the receiving exchange must route to the best protected quote, which a smart order router does.
Rule 610 access: no locked or crossed market from a resting order; routing fee capped at 0.3 cents a share.
Rule 612 sub-penny: minimum tick one cent.
611 against 610 Rule 611 catches orders that execute, Rule 610 orders that rest, and the locking and crossing ban does not reach hidden quotes.
what Rule 611 protects the price and nothing else: a complying order still pays the venue fee, still consumes its liquidity, and still leaks how much remains.
Rules 601, 603 market data: the direct feed.
securities information processor aggregates every exchange's book and trades into the consolidated quotation and consolidated tape, which give the NBBO; nothing trades through it.
flickering quotes exempt from trade-through for up to one second, because the data is older than the market.
NYSE Rule 104 a designated market maker must trade for its own account to keep a fair and orderly market (code 12a).
settlement cycle T+2 for stocks, bonds, mutual funds and municipal securities; T+1 for short-term secured bonds and Treasury bills; T+0 for futures.

3 · Venues and mechanisms

one venue or many Hong Kong runs the HKEX, the United Kingdom the LSE, Japan the TSE; China splits into A and B shares; the US, Canada and India are fragmented.
16 exchanges 13 by 2020. Cboe: BZX, BYX, EDGA, EDGX. ICE: NYSE, Arca, American, National. Nasdaq: Nasdaq, BX, PHLX. Outside: CHX, IEX.
1998 volume NYSE 52%, Nasdaq 40%, other exchanges 8%.
2015 volume FINRA trade reporting facility 20.3%, Nasdaq 15.9%, dark pools 15.1%, NYSE 12.6%, Arca 11.2% (Credit Suisse).
market data revenue the Wall Street Journal of 16 October 2018 reported the SEC blocking higher data fees, the offset to falling trading revenue.
specialist vs dealer NYSE and AMEX run specialist (designated market maker) auctions, whose members may not trade for profit or outbid floor traders; Nasdaq runs a dealer market of registered makers that may.
over-the-counter venues dealer-customer and inter-dealer; request-for-quote platforms from the 2000s; electronic communication networks; the cost is search-and-matching plus counterparty risk, ex-ante and ex-post.
foreign exchange inter-dealer through Reuters/Refinitiv and EBS; about 30 secondary networks, 22 single-dealer platforms, 12 principal trading firms, 2–3 dark pools; volume against volatility $R^2 = 0.6$ over 2010–2021.
Tradition 2019 inter-dealer broker revenue: currencies and rates 43%, securities 32%, commodities 21%, retail foreign exchange 4%.
auction forms first-price, second-price, all-pay; a Dutch auction is a descending clock (first-price), an English one an ascending outcry (second-price); revenue equivalence links them.
gold fixing London, twice a day at 10:30 and 15:00, set by a small panel of members as the reference price, which makes collusion possible.
United States Treasuries primary market runs as a first-price single-sided auction, about a trillion dollars a quarter.
closing price uses fund net asset values, margin, mark-to-market flows, index inclusion, and derivative settlement.
clearing house buyer to every seller and seller to every buyer; guarantees completion, not value; central clearing covers interest-rate swaps and credit default swaps.
central securities depository keeps ownership and custody, completes the trade when one side fails, charges both sides into a buffer, and restricts the failing party's trading.

4 · Hard numbers

Chicago–New York link cable, mid-1980s: 1,000 miles, 14.5 ms; Spread Networks fiber, August 2010: 825 miles, 13.1 ms, the 3 ms saved put at 300 million dollars; McKay and Tradeworx microwave, July 2012: 744 miles, 9 ms, then 731 miles, 8.5 ms; light in vacuum about 5 ms.
May 2024 the Commodity Futures Trading Commission fined J.P. Morgan Securities 100 million dollars over eight years of unsurveilled spoofing.
high-frequency clock microseconds to nanoseconds, a human cannot follow it.
spoofing stack the book on one side with orders meant to be pulled, pull them, then sweep the other side.
quote stuffing Abbott Laboratories, 17 August, 10:07:27: 11,557 orders in one second against a normal 38 a second (Nanex); another burst ran 10,704 then 5,483, all but 14 cancelled within the second.
sniping races the intermarket sweep supplies about 77% of them.
hidden orders 2019 over 18% on NASDAQ, 20% on Frankfurt, 22% on Euronext-Paris.
London, 2003 a random end time for the closing auction, after traders gamed the indicative price.
HKEX closing auction ran from 2008 and was suspended ten months later over suspected manipulation.
NYSE fees take ¢0.21, make ¢0.13; routing ¢0.30, the Rule 610 cap; an inverted schedule pays the taker. A routed make order becomes a taker, and Rule 611 picks price, not fees. Explicit fees overall ≈ 0.05 bp.
sessions pre-market 8:00–9:15, after-market 16:15–18:30; on-open and on-close orders activate at those auctions.

5 · Order vocabulary

four instruction classes time (day, good-till-cancel), execution (immediate-or-cancel, all-or-nothing, fill-or-kill), price (pegged, stop), display (hidden, reserve).
all-or-nothing, fill-or-kill never filled in part; fill-or-kill is all-or-nothing plus immediate-or-cancel.
immediate-or-cancel fills at once, cancels the rest; a RegNMS-compliant form is the NMS IOC at the NYSE.
order lists the NYSE and IEX publish different order-type lists.
stop dormant until the last price touches the stop; a buy triggers upward, a sell downward.
primary peg a buy one tick below the national best bid, a sell one tick above the national best offer; hidden.
midpoint peg priced at the national best bid and offer (NBBO) midpoint.
hidden order shows part of its size; ranks behind displayed orders at the same price.
dark pool an alternative trading system, fully hidden, executing at the NBBO midpoint by pro-rata crossing: $P = \min\left(1, \frac{n_s}{n_b}\right)$.
intermarket sweep the broker merges all quote feeds and releases one child order per venue in the same instant; exempt from Rule 611.
do-not-ship the venue's router is switched off, trading price protection away; the limit form cancels whatever would lock or cross.
non-routable intermarket sweep and do-not-ship, each in immediate-or-cancel and limit form: four types.
splitting an order pro-rata allocation is the most popular routing pattern, then time priority, then round robin.

6 · What NYSE order types do (Li, Ye & Zheng 2022)

type	trades	size	fill	take	route	make
plain market	6.36M	278.96	100%	66.34	33.64	0
plain IOC	4.20M	268.63	19.48%	98.07	1.93	0
ISO	25.41M	244.30	22.48%	100	0	0
DNS IOC	14.65M	307.09	31.86%	100	0	0
day limit	43.36M	220.50	2.71%	29.78	14.28	55.94
DNS limit	44.24M	182.38	3.72%	11.43	0	88.57

routing refused 57% of New York Stock Exchange orders refuse the routing Regulation NMS requires.
first responders 495 a day against 1,652 trades, a non-marketable limit refilling a consumed price within 0.1 s; 75.6% do-not-ship limit, 24.4% day limit.
cancellation race a do-not-ship limit escapes a sniping event 42% of the time; a routable limit almost never does.
race winners only the plain IOC announces itself, and it alone loses more races than it wins (13.42 against 17.37); the ISO wins 57.26% of 140.71, the DNS IOC 37.28% of 91.63.
weighted price contribution the order types partition the price move and sum to 100%; the non-routable types contribute more than the whole move, so they trade on new information.
price improvement 26.99% of routed market orders fill at the arrival quote, none of the routed plain IOC; 77.49% of day limit and 79.51% of reserve orders get no improvement.

7 · Data traps

bid-ask bounce execution prices zigzag between bid and ask with no change in value, so a computed variance reads mechanics as news; Roll (1984) measures it.
inside the spread internalized orders (NYSE floor traders), hidden orders, and midpoint pegs at IEX; the terms are not in limit-order-book (LOB) data.
outside the spread an intermarket sweep prints worse than the NBBO and the securities information processor records it; negotiated blocks trade "upstairs" through NYSE specialists at prices far off the NBBO.
VWAP prints a block sold by negotiation at 3:30 pm at the volume-weighted average price prints above the best ask, so price data alone reads it as a buy.
double counting where both brokers report, the consolidated tape counts each trade twice.
LOB data records winners only: a failed cancel and a losing immediate-or-cancel order are absent.
message data every attempt, time-stamped at the gateway; London Stock Exchange data are available under Section 165 of the Financial Services and Markets Act; TAQ and MIDAS are the consolidated LOB sets.

8 · Platform and post-trade

circuit firewall, gateways (validating), sequencer (round-robin, fixing arrival order), matching engine (serial, price-time), then distribution server and market data processor.
disclosed platforms the London Stock Exchange, Deutsche Börse, and the Chicago Mercantile Exchange publish their design; most others do not.
speed bump a venue that sells speed services, such as a direct data feed, while deliberately slowing incoming orders.
delivery versus payment each leg is released only once the other is in place, so a failure on one side leaves the other side whole.
physical delivery in-kind commodities and futures add storage and financing cost, and markets for temporary storage and substitutes.
participants broker earns a fee and keeps no position, dealer is the other side and earns the spread, broker-dealers do both.
investors retail trade through brokers; institutions hold only vetted securities, so index inclusion shifts the investor base, the index effect.
speculators and arbitrageurs speculators take a directional view; arbitrageurs trade a price gap between two markets or instruments.
hedgers, producers, government hedgers offload other risk and are the "noise" of theory; producers hold private supply information; governments act through open market operations and quantitative easing.

9 · Course vocabulary

two liquidity senses market liquidity is how cheaply an asset trades; funding liquidity is the financing a trader needs.
market quality two pillars: liquidity, and price discovery (price efficiency).
three perspectives positive (what happens), equilibrium (what each player does, what results), normative (how to design and regulate).
market-based evaluation welfare is unobservable, so structure is judged on prices and market metrics through the efficient-market hypothesis.
efficient price $p^* = E[v \mid \text{available information}]$; deviations come from absent informed trading, information frictions, and structural frictions.
trading needs private value, information or belief, and non-fundamental needs such as needing cash.
order attributes price, quantity, direction, plus priority and speed, execution quality and risk, transparency.
locked and crossed the best bid equals the best ask (locked) or exceeds it (crossed); impossible on one book, routine across venues and in limit-order-book data.
marketable limit order priced at or through the far quote, so it takes liquidity and, once filled, is indistinguishable from a market order.
near and far side a buyer's near side is the bid and its far side the ask; a seller's is the reverse.
mid-point quote the average of the best bid and the best ask, used as the fair-price reference and never traded at.
auction phases call phase (indicative price and quantity only, ending at a random time), price determination (maximizes volume), market balancing (makers absorb the excess).
Walras an auctioneer calls a tentative price and moves it until demand equals supply; with frictions the clearing problem is NP-hard and several prices can clear.

10 · Formulas (notes 1–5)

spread $a - b$ quoted; relative spread $\frac{a-b}{m}$; a buyer pays $a - m = \frac{a-b}{2}$ above the mid, a seller the same below.
mid price $m = \frac{a+b}{2}$, the reference the pegs and the dark pools track.
effective spread $2|p - m|$ for a trade at p; price improvement is $a - p$ on a buy, $p - b$ on a sell.
fee arithmetic the venue keeps $f_{\text{take}} - f_{\text{make}}$; Rule 610 caps $f_{\text{route}} \le 0.3$ cents, and 0.3 cents on a 100-dollar share is 0.3 bp.
tick value 1 cent on a 100-dollar share is 1 bp; the grid is $p_0 + k$ cents, so 101.255 is not a price.
dark pool fill $P = \min\left(1, \frac{n_s}{n_b}\right)$; an informed buyer faces $\min\left(1, \frac{n_s^U}{n_b^U + n_b^I}\right)$.
light in fiber about 2×10^8 m/s, so 5 microseconds a kilometre; New York to Chicago is roughly 1,200 km one way.
order arrivals Poisson with rate λ, so $\Pr(\text{none in } \Delta t) = e^{-\lambda \Delta t}$ and the mean wait is $\frac{1}{\lambda}$ (Garman).
a race of equals with rival orders of equal speed each wins with probability $\frac{1}{k}$, so speed is a winner-takes-all investment.
funding a settlement two extra days of cash cost $2\frac{r}{365}$ of the notional at rate r, the price of $T + 2$ over $T + 0$.
auction price one price clears the largest volume, where the demand and supply curves cross; Walras moves the tentative price up while $D > S$ and down while $D < S$.

11 · Information and spread formulas (notes 6–9, taught before this midterm)

conditional probability $\Pr(A|B) = \Pr\frac{A \cap B}{\Pr}(B)$.
Bayes worked prior 0.6 infected, $\Pr(-|I) = 0.05$, $\Pr(-|N) = 0.95$: posterior odds $\frac{1.5}{19}$, so $\Pr(I|-) = \frac{3}{41}$, about 7%.
total probability $\Pr(A) = \sum_i \Pr(B_i)\Pr(A|B_i)$ over a partition B_i.
Bayes $\Pr(A|B) = \Pr(B|A)\Pr\frac{A}{\Pr}(B)$; posterior odds = prior odds × likelihood ratio.
binary signal prior $\frac{1}{2}$ and precision δ give $\Pr(V_H|S_H) = \delta$, posterior odds $\frac{\delta}{1-\delta}$.
normal update $E[v|s] = \frac{\tau}{\tau+\tau_e}\mu + \frac{\tau_e}{\tau+\tau_e}s$, with $\tau = \frac{1}{\sigma^2}$ and $\tau_e = \frac{1}{\sigma_e^2}$.
conditioning on y $E[x|y] = \mu_x + \rho\left(\frac{\sigma_x}{\sigma_y}\right)(y - \mu_y)$ and $\text{Var}(x|y) = \sigma_x^2(1 - \rho^2)$, the general form the normal update is a case of.
posterior variance $\text{Var}(v|s) = \frac{1}{\tau+\tau_e} = \sigma^2\frac{\sigma_e^2}{\sigma^2+\sigma_e^2}$; independence is what lets the precisions add.
signal response $E[v|s] = \mu + \rho^2(s - \mu)$, the same weight written as the squared correlation $\rho^2 = \frac{\sigma^2}{\sigma^2+\sigma_e^2}$, which is the regression R^2.
signal informativeness $\eta_s = \text{Var}\frac{v}{\text{Var}}(v|s) = 1 + \frac{\tau_e}{\tau} = \frac{1}{1-\rho^2} \ge 1$, so $\text{Var}(v|s) = \text{Var}\frac{v}{\eta_s}$.
price efficiency $\eta_p = \text{Var}\frac{v|s}{\text{Var}}(v|p) = \frac{\tau_v+\tau_p}{\tau_v+\tau_e} \le 1$, a different ratio: the price against the insider, not the prior. It reaches 1 only when $\sigma_z = 0$.
price precision $\tau_p = \frac{1}{\sigma_e^2 + \left(\frac{\sigma_z}{n_I\tau_e}\right)^2}$, so it rises with n_I and with τ_e but never reaches τ_e.
the two etas opposite directions: $\eta_s \ge 1$ says how much a signal adds and equals 1 when it adds nothing, while $\eta_p \le 1$ says how much of the insider the price carries and equals 1 when the price knows everything.
standardized signal $\frac{s-\mu}{\sigma}$ leaves the informativeness unchanged, so signals in different units can be ranked by one number.
demand in equilibrium $x = \frac{E[v|\text{info}]-p}{\gamma\,\text{Var}(v|\text{info})}$ and $p^* = \frac{n_I\tau_e s - X}{n_I(\tau_v+\tau_e)+n_U\tau_v}$, so supply X lowers the price and the signal raises it.
price as signal $p = \alpha + \beta s + \gamma z$ with $\beta = n_I\frac{\tau_e}{n_I\tau_e+N\tau_v}$ and $\frac{\gamma}{\beta} = \frac{1}{n_I\tau_e}$.
price efficiency $\tau_p = \frac{1}{\text{Var}}(v|p) < \tau_e$ and $\eta < 1$ once noise trades; at $z = 0$ the price reveals s exactly and the informed pay for nothing.
Glosten–Milgrom quote $\Pr(\text{buy}|v = +-\sigma) = \frac{1+-\mu}{2}$, so $A^* = E[v|\text{buy}] = \mu\sigma$ and $B^* = -\mu\sigma$.
spread from information $S = A^* - B^* = 2\mu\sigma$; the maker loses $(1 - \mu)\sigma$ on each informed trade and gains $\mu\sigma$ on each noise trade, so the transfer is $\mu(1 - \mu)\sigma$.
pure adverse selection in Glosten–Milgrom no inventory and no processing cost enters, so the whole spread is information.
Roll price $p_t = m_t + \left(\frac{S}{2}\right)d_t$ with $d_t = +-1$ equally likely and $m_t = m_{t-1} + e_t$, e_t i.i.d. normal.
Roll changes $\Delta p_t = \left(\frac{S}{2}\right)(d_t - d_{t-1}) + e_t$: only a switch of side moves the print, and then by S.
Roll moments $E[\Delta p_t] = 0$ and $\text{Var}(\Delta p_t) = \frac{S^2}{2} + \sigma_e^2$, so the bounce alone is a floor under measured variance.
Roll estimator $\text{Cov}(\Delta p_t, \Delta p_{t-1}) = -\frac{S^2}{4}$, hence $S = 2\sqrt{-\text{Cov}(\Delta p_t, \Delta p_{t-1})}$ and $\sigma_e^2 = \text{Var}(\Delta p) - \frac{S^2}{2}$.
bounce share $100 \times \frac{\frac{S^2}{2}}{\text{Var}}(\Delta p_t)$ per cent of measured variance is trading cost, not news.
no-trade theorem identical information rules out trade at every price; differing signals leave the open interval between the two valuations, and full common knowledge closes it.
no-trade worked with $V_H = 120$, $V_L = 0$ and precision $\frac{2}{3}$ the two valuations are 80 and 40, so the trade interval is the open span between them.
complete markets the theorem needs complete markets and common knowledge of rationality as well as common priors; drop any one and trade returns.
tied races a race that q orders tie is split $\frac{1}{q}$ each, then averaged at the stock-day level (Li, Ye & Zheng).
Grossman–Stiglitz demand $U(x) = E[x(v - p)] - \left(\frac{\gamma}{2}\right)\text{Var}(xv)$, so $x = \frac{E[v|\text{info}]-p}{\gamma\,\text{Var}(v|\text{info})}$: an edge over a risk.
strong-form efficiency impossible when information is costly, since it needs $z = 0$ and without noise nobody pays to learn; the price is semi-strong at best.

12 · Weighted price contribution by order type (Li, Ye & Zheng 2022)

type	volume share	WPC
plain market	6.77%	−4.56%
plain IOC	6.36%	3.86%
ISO	35.69%	90.32%
DNS IOC	25.87%	27.10%
day limit	16.37%	−13.78%
DNS limit	5.30%	−2.34%
reserve limit	3.43%	−0.95%
DNS reserve limit	0.21%	0.35%

sums the non-routable types reach 115.43% and the routable ones −15.43%, so the types that refuse routing produce the whole move and the excess.
which type moves the price the ISO, on a third of the taking volume; the largest negative is the day limit order.
waiting beats arriving late the DNS IOC and the DNS limit share a routing instruction but score +27.10% and −2.34%: what separates them is whether the order waits on the book or executes on arrival.